

PIYUSH BIPIN VARULE

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Summary

Third-year B.Tech student in Artificial Intelligence and Machine Learning (CGPA: 9.14) with hands-on experience developing AI applications, computer vision systems, full-stack web platforms, and intelligent automation solutions. Built projects using Python, React, TypeScript, OpenCV, PyTorch, FastAPI, and Docker, including AI-powered healthcare, computer vision, and offline-first data validation systems. Passionate about applying AI to solve real-world engineering problems through scalable software solutions.

Experience

May 2025 - Jun 2025

Technical Content & Research Intern

Amigoes Foundation

- Researched technical and awareness topics to produce SEO-optimized digital content across multiple campaigns.
- Improved content structure and readability by collaborating with editors.
- Created documentation and structured information from multiple verified sources while consistently meeting project deadlines.
- Strengthened research, analytical thinking, and technical communication skills applicable to software documentation.

Mar 2026 - Apr 2026

Freelance AI/ML Developer

Self-Employed (Freelance)

- Developed an AI-powered computer vision application using Python, OpenCV, Open3D, and PyTorch to generate 3D foot models from smartphone images.
- Implemented 4 core modules: image acquisition, 3D reconstruction, automated measurement extraction, and visualization.
- Integrated computer vision and deep learning techniques to support custom footwear, orthopedic assessment, and digital health applications.
- Built export functionality for sharing measurements and generated 3D models.

Education

2024 - 2028

Pimpri Chinchwad University, Pune

Undergraduate •Bachelor Of Technology

Percentage - 91.4% •CGPA - 9.14

Currently pursuing a Bachelor of Technology in Artificial Intelligence and Machine Learning with a strong academic record of 9.14 CGPA. Gaining hands-on experience in Machine Learning, Full-Stack Development, Data Structures & Algorithms, and AI-based application development through academic projects, hackathons, and technical certifications.

Certification

Oracle AI Foundations Associate

[Certificate Link](#)

Oracle Generative AI Professional

[Certificate Link](#)

Introduction to Generative AI

[Certificate Link](#)

Introduction to Operating Systems Specialization

[Certificate Link](#)

Foundations of Cybersecurity

[Certificate Link](#)

5-Day AI Agents Intensive Course with Google

[Certificate Link](#)

Skills

- | | | | |
|--------------|---------------------------|-------------------------------------|----------------|
| • JavaScript | • CPP | • Natural Language Processing (NLP) | • Tailwind CSS |
| • Html5 | • java | • Generative AI | • MongoDB |
| • Python | • Artificial Intelligence | • Prompt Engineering | • VS Code |
| • DSA | • bootstrap | • Data Analysis | • |
| • C | • CSS | • NumPy | |

Projects

Formguard-Insight

[Project Link](#)

- Developed an offline-first AI-assisted validation platform supporting 3 form types (Student, Hospital, and Store).
- Implemented anomaly detection, dynamic validation rules, role-based approval workflows, and analytics dashboards.
- Enabled offline data storage using IndexedDB with export support for CSV, JSON, and PDF.
- Built a responsive frontend using React, TypeScript, Tailwind CSS, and Chart.js.

MediBuddy v2 - AI Multi-Agent Medication Assistant

[Project Link](#)

- Designed an AI multi-agent system comprising reminder, interaction, adherence tracking, and analytics agents.
- Implemented rule-based coordination and machine learning techniques for personalized medication reminders.
- Built adherence monitoring workflows that generated user-specific medication reports.
- Designed the solution for healthcare scenarios requiring intelligent automation and patient engagement.

DigiFoot – AI 3D Foot Scanning System

- Developed a computer vision-based system using OpenCV and Open3D for real-time 3D foot reconstruction from smartphone images.
- Implemented deep learning models with PyTorch to enhance accuracy in measurement extraction and model refinement.
- Built features for 3D visualization, automated dimension analysis, and export functionality for practical applications.
- Designed the solution for use cases in custom footwear, orthopedic assessment, and digital health monitoring.

Blue Insights – AI Marine Data Platform

[Project Link](#)

- Developed a unified analytics platform integrating 3 major marine data sources: oceanographic, fisheries, and satellite datasets.
- Built Docker-based data pipelines for scalable processing and cloud-ready deployment.
- Applied machine learning techniques for fish stock prediction, algal bloom analysis, and biodiversity hotspot identification.
- Designed the platform to support India's Blue Economy initiative and SDG-14.

Responsibility

AI & ML Student Volunteer Club, Pimpri Chinchwad University

Technical Team Member

Organized technical workshops and coding sessions.

Assisted in hackathons and AI-related events.

Collaborated with team members to promote technical learning.